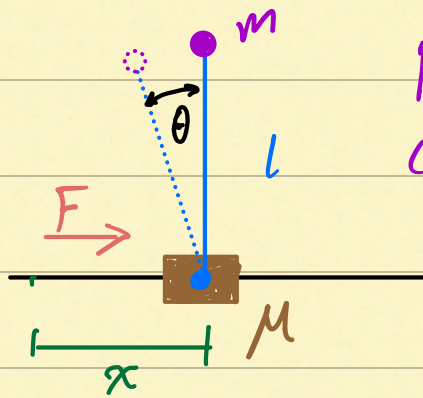




pole: $x_p = x - \sin\theta l$ $y_p = \cos\theta l$

cart: $x_c = x$ $y_c = 0$

$$\dot{x}_p = \dot{x} - \cos\theta \dot{\theta} l \quad \dot{y}_p = -\sin\theta \dot{\theta} l$$

$$\dot{x}_c = \dot{x} \quad \dot{y}_c = 0$$

$$\begin{aligned} v_p^2 &= \dot{x}_p^2 + \dot{y}_p^2 = \dot{x}^2 - 2\dot{x}\cos\theta\dot{\theta}l + \cos^2\theta\dot{\theta}^2 l^2 + \sin^2\theta\dot{\theta}^2 l^2 \\ &= \dot{x}^2 - 2\dot{x}\cos\theta\dot{\theta}l + \dot{\theta}^2 l^2 \end{aligned}$$

$$L = T - V$$

T — kinetic energy

V — potential energy

For whole system

$$L = \frac{1}{2} m v_p^2 + \frac{1}{2} M v_c^2 - mgy_p$$

$$= \frac{1}{2} m (\dot{x}^2 - 2\dot{x}\cos\theta\dot{\theta}l + \dot{\theta}^2 l^2) + \frac{1}{2} M \dot{x}^2 - mgl\cos\theta$$

$$= \frac{1}{2} (m+M) \dot{x}^2 - m\dot{x}\cos\theta\dot{\theta}l + \frac{1}{2} m \dot{\theta}^2 l^2 - mgl\cos\theta$$

For x Euler-Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = Q_i$$

$$\frac{\partial L}{\partial \dot{x}} = (m+M)\dot{x} - m \cos \theta \dot{\theta} l \quad \frac{\partial L}{\partial x} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = (m+M)\ddot{x} + m \sin \theta \dot{\theta}^2 l - m \cos \theta \ddot{\theta} l$$

$$\text{So: } (m+M)\ddot{x} + m \sin \theta \dot{\theta}^2 l - m \cos \theta \ddot{\theta} l = F$$

$$(m+M)\ddot{x} + ml (\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta}) = F$$

For θ

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q_i$$

$$\frac{\partial L}{\partial \dot{\theta}} = -m \dot{x} \cos \theta l + m \dot{\theta} l^2$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = -m \ddot{x} \cos \theta l + m \dot{x} \sin \theta \dot{\theta} l + m \ddot{\theta} l^2$$

$$\frac{\partial L}{\partial \theta} = m \dot{x} \sin \theta \dot{\theta} l + mg \sin \theta l$$

$$\begin{aligned}
 \text{So: } & -m\ddot{x}\cos\theta l + m\dot{x}\sin\theta\dot{\theta}l + m\ddot{\theta}l^2 - m\dot{x}\sin\theta\dot{\theta}^2l - mg\sin\theta l = 0 \\
 & -m\ddot{x}\cos\theta l + \dot{x}(m\sin\theta\dot{\theta}l - m\sin\theta\dot{\theta}l) - mg\sin\theta l + m\ddot{\theta}l^2 = 0 \\
 & -m\ddot{x}\cos\theta l - mg\sin\theta l + m\ddot{\theta}l^2 = 0 \\
 & \ddot{x}\cos\theta + g\sin\theta - \ddot{\theta}l = 0
 \end{aligned}$$

system dynamic:

$$(m+M)\ddot{x} + ml(\sin\theta\dot{\theta}^2 - \cos\theta\ddot{\theta}) = F \quad (1)$$

$$\ddot{x}\cos\theta + g\sin\theta - \ddot{\theta}l = 0 \quad (2)$$

$$\underbrace{\begin{bmatrix} m+M & -ml\cos\theta \\ \cos\theta & -l \end{bmatrix}}_A \underbrace{\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix}}_B = \underbrace{\begin{bmatrix} F - ml\sin\theta\dot{\theta}^2 \\ -g\sin\theta \end{bmatrix}}_B$$

$$\begin{aligned}
 A^{-1} &= \frac{1}{|A|} A^* = \frac{1}{-(m+M)l + ml\cos^2\theta} \begin{bmatrix} -l & m\cos\theta \\ -\cos\theta & m+M \end{bmatrix} \\
 &= \frac{1}{-M(-ml(1-\cos^2\theta))} \begin{bmatrix} -l & m\cos\theta \\ -\cos\theta & m+M \end{bmatrix} \\
 &= \frac{1}{-(M+ml\sin^2\theta)} \begin{bmatrix} -l & m\cos\theta \\ -\cos\theta & m+M \end{bmatrix}
 \end{aligned}$$

$$\text{set } D(\theta) = M + m \sin^2 \theta$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = A^{-1} \cdot B = \frac{1}{-D(\theta)} \begin{bmatrix} -F + m(\sin \theta \dot{\theta}^2 - mg \cos \theta \sin \theta) \\ -\cos \theta F + m(\sin \theta \cos \theta \dot{\theta}^2 - g \sin \theta (m + M)) \end{bmatrix}$$

Transfer to State Space

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$